ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 2 style (design-related), built around one case study: an automated water bottling line. Section A: objective questions (30) · Section B: line layout, bottlenecks and filling sequence, conveyor speed, pulley, gearbox, torque and power sizing, VSDs, PLC I/O, reject timing and HMI alarms, maintenance planning and condition monitoring (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A capping machine screws … onto bottles.

A labels
B bottles
C pallets
D caps

1.1.2 On a bottling line, a photoelectric sensor is mostly used to …

A heat the bottles.
B detect the bottles.
C fill the bottles.
D clean the bottles.

1.1.3 A star wheel is used to …

A space and transfer bottles.
B paint bottles.
C seal boxes.
D weigh trucks.

1.1.4 A line speed of 60 bottles per minute is equal to … bottles per hour.

A 60
B 600
C 3 600
D 36 000

1.1.5 A variable speed drive (VSD) controls the speed of an AC motor by changing the …

A colour of the motor.
B supply frequency.
C number of bottles.
D length of the belt.

1.1.6 A recipe on the HMI stores …

A operator passwords.
B motor power ratings.
C maintenance dates.
D the settings for each product, such as the fill volume.

1.1.7 An air-blast reject unit removes …

A faulty bottles from the line.
B every bottle.
C only the labels.
D water from the floor.

1.1.8 The bottleneck of a production line is the …

A fastest machine.
B first machine.
C slowest machine, which limits the output.
D longest conveyor.

1.1.9 A flow meter on the filler measures the …

A temperature of the bottle.
B volume of liquid passing through.
C bottle height.
D motor current.

1.1.10 Lockout-tagout is carried out …

A before maintenance, so that the machine cannot start.
B after every bottle.
C to speed up the line.
D to fill the bottles.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A sequence chart shows the order in which the actuators operate.

1.2.2 Bottles should be spaced evenly before they enter the filler.

1.2.3 The output of a line is set by its fastest machine.

1.2.4 A level or weight check can detect under-filled bottles.

1.2.5 Vibration monitoring can detect worn bearings early.

1.2.6 The speed of a conveyor belt does not depend on the diameter of the drive pulley.

1.2.7 Emergency-stop buttons must be easy to reach along the whole line.

1.2.8 A maintenance plan should only be drawn up after a breakdown.

1.2.9 Food-grade lubricants are used where they might touch the product.

1.2.10 A VSD always runs the motor at full speed.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Machine that fills bottles with a set volume	A	recipe
1.3.2	Machine that fits and tightens the caps	B	lockout-tagout
1.3.3	Slowest stage, which limits the output of the line	C	filler
1.3.4	Controls the speed of an AC motor by changing the frequency	D	cycle time
1.3.5	Stored set of machine settings for one product	E	VSD
1.3.6	Device that removes faulty bottles from the conveyor	F	pitch
1.3.7	Distance from the centre of one bottle to the next on the conveyor	G	capper
1.3.8	Measuring vibration, heat or oil to judge the health of a machine	H	condition monitoring
1.3.9	Isolating a machine and locking it so that it cannot start during maintenance	I	bottleneck
1.3.10	Time taken for one machine to complete its work on one bottle	J	reject unit

(10)

TOTAL SECTION A: [30]

SECTION B

CASE STUDY. A factory is designing an automated line that bottles 500 ml water. Empty bottles move on a conveyor through a filler, a capper, a labeller and a camera inspection station with a reject unit, and are then packed into boxes. The line must produce at least 3 600 bottles per hour. A PLC controls the line and the operators use an HMI.

QUESTION 2: LINE DESIGN AND SEQUENCE

2.1 Draw a labelled block layout of the bottling line, showing the machines in the correct order, the conveyors and at least TWO sensors. (5)

2.2 The machines can handle the following numbers of bottles per minute: filler 70, capper 65, labeller 80 and packer 75. Identify the bottleneck, calculate the output of the line per hour and state whether the line meets the target. (4)

2.3 At the filler, a stopper cylinder holds each bottle under the nozzle, a nozzle cylinder lowers the nozzle into the bottle, and a valve fills the bottle. List the steps of the filling sequence in the correct order, and name the sensor or signal that confirms EACH step before the next one starts. (6)

2.4 The filling valve delivers water at 0,25 L/s. Calculate the time needed to fill one bottle. (3)

[18]

QUESTION 3: DRIVE SIZING

The bottles are carried on the conveyor at a pitch of 100 mm and the conveyor must move 65 bottles per minute. The drive pulley has a diameter of 100 mm, and the belt pull at the drive pulley is 300 N.

3.1 Calculate the belt speed in m/s. (3)

3.2 Calculate the speed of the drive pulley in rpm. (3)

3.3 The motor runs at 1 400 rpm. Calculate the gearbox ratio needed. (2)

3.4 Calculate the torque and the power needed at the drive pulley. (4)

3.5 The gearbox efficiency is 80% and the motor efficiency is 85%. Calculate the electrical input power of the motor. (3)

3.6 Explain why a VSD is used to drive the conveyor motor. (3)

[18]

QUESTION 4: PLC CONTROL AND INSPECTION

4.1 Draw up an I/O allocation table for the capper and the reject station, with at least FOUR inputs and THREE outputs. (7)

4.2 The camera finds a faulty bottle, but the reject unit is 5 bottles further down the conveyor. Explain how the PLC makes sure that the correct bottle is blown off. (4)

4.3 The reject valve Y5 must switch on for 0,2 s when the "bottle at reject" sensor X6 is on and the fault memory M1 is set. Draw the ladder rungs, using an on-delay timer T1. (4)

4.4 Name THREE alarms that should be shown on the HMI of the line. (3)

[18]

QUESTION 5: MAINTENANCE AND CONDITION MONITORING

5.1 Draw up a maintenance plan for FOUR items on the bottling line. For each item, give the maintenance task and how often it must be done. (8)

5.2 Name THREE condition-monitoring methods and state what EACH one detects. (6)

5.3 Explain why hygiene and food-grade lubricants are important when maintaining this line. (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER